

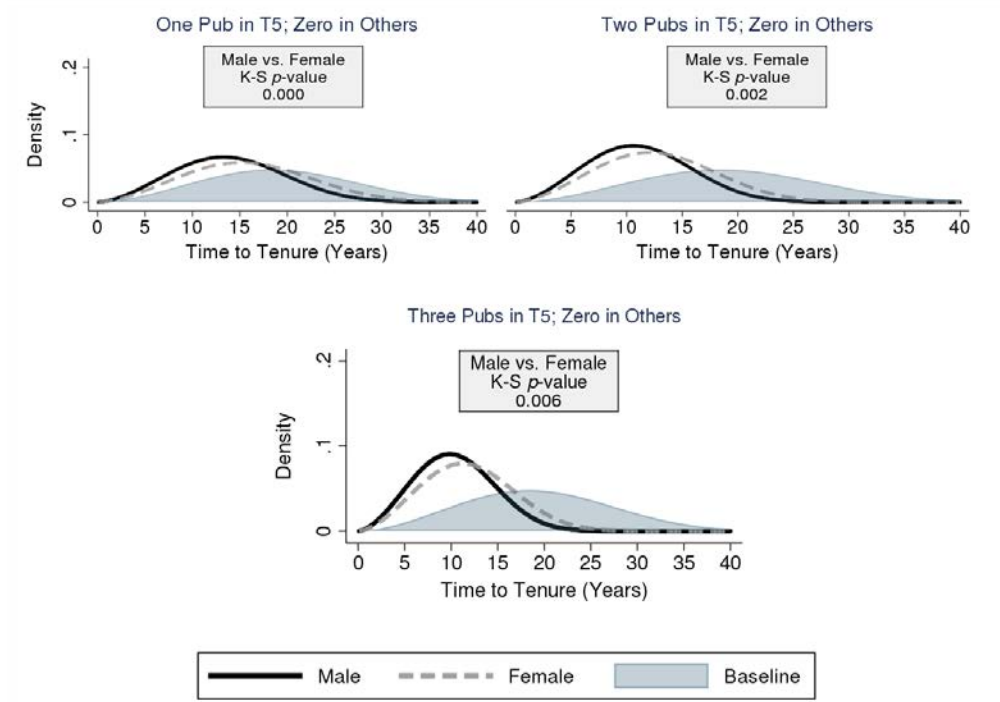
heterogeneous across the different department rank groupings. The first T5 publication is estimated to have a significant impact on the hazard of tenure for only those individuals employed by departments ranked 21–35. Further, the magnitude of impact associated with the first T5 publication is higher for these departments compared to higher ranked departments. Conversely, the estimates associated with two or three T5 publications is only significant for the top 20 departments, and the magnitude of impact associated with these publications is higher for individuals hired by the top 10 departments compared to individuals in departments ranked 11–20. Online Appendix Section 4.2 presents analogous rank-specific estimates for the non-T5 journal categories. T5 publications are estimated to have a larger impact on the hazard of tenure relative to non-T5 publications, across all department rank groupings.

2.4 Heterogeneity in the Probability and Rate of Receiving Tenure By Gender

We next investigate gender differences in time-to-tenure and in the probability of receiving tenure. Duration analyses in Section 2.4.1 reveal that male faculty enjoy shorter times-to-tenure than their female colleagues. We investigate the source of this discrepancy by checking whether differences in time-to-tenure stem from differential returns to publication by gender. Our analysis shows that T5 publications are indeed associated with greater reductions in time-to-tenure for male faculty. We are careful not to interpret these estimates as evidence of discrimination since we lack the data required to control for the confounding effect of fertility on female time-to-tenure. While gender differences exist for time-to-tenure, differences by gender are not present in our analyses of the probability of receiving tenure (both during the first spell and by the 7th year of tenure-track employment) in Section 2.4.2. Our results thus suggest that female faculty take longer to receive tenure but are not less likely to eventually receive tenure.

2.4.1 Heterogeneity in Time-to-Tenure

Figure 9: Densities of Time-to-Tenure (Weibull Distribution), by Gender (Publication Rewards Held Constant across Genders)



Note: This figure plots conditional densities of time-to-tenure given different levels of publications in the T5 journals and gender. Densities of time-to-tenure are predicted using parameter estimates obtained by estimating Text-Appendix Equation TA-13 without interacting the publication parameters with gender. Conditional densities of time-to-tenure given gender g , x number of T5s, and 0 non-T5s is given by: $f(t \mid \#T5 = x, \#nonT5 = 0, \text{Gender} = g, \mathbf{X}) = h(t \mid \#T5 = x, \#nonT5 = 0, \text{Gender} = g, \mathbf{X}) \times S(t \mid \#T5 = x, \#nonT5 = 0, \text{Gender} = g, \mathbf{X})$ where $h(t \mid \cdot)$ and $S(t \mid \cdot)$ give the conditional hazard and survivor rates at t respectively. The titles for the 3 subplots in Figure 9 lists the conditioning used for the publication variables. “2 Pubs in T5; 0 in Others” gives the condition: $\#T5 = 2, \#nonT5 = 0$. The conditioning on gender is given by the legend which denotes whether the plot is associated with males or females. The conditioning on $\mathbf{X}$ is left implicit. Taken together, the black density for the plot labelled “2 Pubs in T5; 0 in Others” plots the following density function: $f(t \mid \#T5 = 2, \#nonT5 = 0, \text{Gender} = \text{Male}, \mathbf{X})$. Plots for females and other quantity of T5s are analogously defined. The blue shaded region in each plot represents the conditional density of time-to-tenure for females given zero publications in all outlets journal. Each plot also presents p -values obtained from Kolmogorov-Smirnov tests between the Male and Female distributions.

This section investigates heterogeneity in time-to-tenure and tenure rates by gender. Estimation of the baseline hazard yields an estimated hazard ratio for the gender indicator (denoting that the subject is male) that ranges between 1.43 and 1.44 depending on the assumption made about unobserved heterogeneity (see Online Appendix Table O-A17).

The ratio of 1.44 for men indicates that male faculty have a rate of time-to-tenure that is 44% greater than those faced by their female colleagues, once differences in the number of publications and the vector of time-variant and -invariant controls $\mathbf{X}$ and $\overline{\mathbf{C}}$ are accounted for. Analyzing data for an older sample of individuals who were employed as economists in 1989, [Kahn \(1993\)](#) finds that men enjoy hazards of tenure that are 56% higher than that for women.⁴³ The difference in hazard rates translates into differences in time-to-tenure. Figure 9 plots gender-specific densities of time-to-tenure associated with publishing one to three T5 publications (see Online Appendix Figure O-A24 for non-parametric Kaplan Meier plots of survival probabilities by gender and number of T5 publications). The densities for females exhibit a rightward shift relative to their male counterparts. Kolmogorov-Smirnov tests reported in the figure reject the null hypothesis of distributional equality across genders at the 5% level for each level of T5 publication.

Given the statistical significance of the gender indicator, we next investigate differences in rewards associated with T5 publications by gender. We explore heterogeneity in rewards to publication by interacting the publication variables in Equation 1 with an indicator for gender. Online Appendix Figures O-A21–O-A23 present gender-specific tenure hazards associated with different levels of publishing in different journal categories. Figure O-A21 plots hazards associated with the first three T5 publications, by gender. Females are estimated to have higher hazard rates to tenure for the first T5 publication. However, the estimate associated with the first T5 is only statistically significant for male faculty at the 5% level. Males are estimated to have markedly higher hazard rates than females for the second and third T5 publication. The hazard rate associated with two T5 publications is 56% higher for males than for females. The hazard rate associated with three T5 publications is 86% higher for males. The hazard rate estimates for the second and third T5 publications are only statistically significant for male faculty. These gender differences in hazard rates suggest that male faculty reap greater rewards for T5 publication—the same

⁴³Our estimate is not directly comparable with that of [Kahn \(1993\)](#) because we adopt different model specifications and use a richer control set (our data collection procedure yields richer bibliographic data).

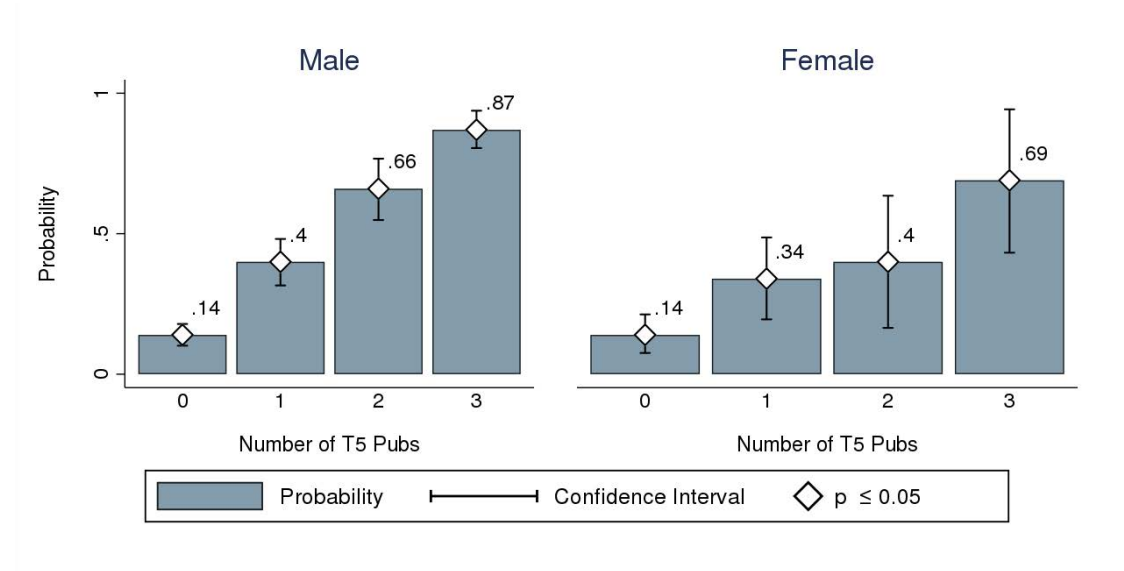
quantity of T5 publications is associated with greater reductions in time-to-tenure for male faculty compared to their female counterparts. Gender differences in T5 rewards are not attributable to gender differences in the quality of T5 articles. An inter-gender comparison of citation distributions for solo-authored T5 articles reveals that citations to T5 articles are not statistically significantly different across genders (see Online Appendix Section 7.6 for details). However, point estimates for female faculty are more imprecisely determined than those for males due to the relatively small sample of female faculty.⁴⁴

The difference in tenure hazards and time-to-tenure across genders suggests that female faculty receive lower and possibly more uncertain rewards than their male counterparts for the same level of publications. It is unclear how much of the slower female rate to tenure is accounted for by fertility and differences in the take-up rate of parental leave (pregnancy-related leave) between male and female faculty. Antecol et al. (2018) present evidence that gender neutral tenure clock-stopping policies have differential impacts on rates of tenure receipt by gender. The adoption of gender-neutral tenure clock-stopping policies is found to be associated with significant reductions in the rate of tenure receipt for female faculty in economics but not for their male counterparts. Ginther and Kahn (2004) find that 10 years post-PhD, female faculty in economics with children are statistically significantly less likely to receive tenure than female faculty without children. Such differences do not exist between male faculty with and without children. The existence of such discrepancies across male and female faculty suggest that childbearing and rearing have important differential impacts on time-to-tenure by gender, and that such differences would be exacerbated by tenure clock-stopping policies. Unfortunately, we lack the requisite data to make the appropriate adjustment to female exposure sets.

⁴⁴The sample size is small for two reasons: (i) there are fewer females than males in academic economics (Scott and Siegfried (2018) report that women accounted for 21.7%–26.6% of assistant and associate professor positions in the 2017-2018 academic year across 103 PhD-granting institutions in the U.S); and (ii) women who publish 3 or more T5 articles are much fewer in number.

2.4.2 Heterogeneity in the Probability of Receiving Tenure

Figure 10: Conditional Probabilities of Receiving Tenure During the First Spell of Tenure-Track Employment Given T5 Publications and Gender



Note: This figure plots conditional probabilities of receiving tenure during the first spell of tenure-track employment, given the quantity of T5 publications and gender. The probabilities are estimated as proportions of individuals within each gender-T5 quantity cell who received tenure during the first spell of tenure-track employment.

Figure 10 plots raw probabilities of tenure given gender and number of T5 publications.⁴⁵ The probabilities are lower for females at all levels of T5 publication. This result suggests that females might reap lower rewards (in terms of the probability of receiving tenure) than males for the same number of T5 publications. Although Figure 10 indicates that tenure probabilities vary by gender given the same number of T5 publications, these gender differences disappear when we estimate logit models that include an indicator for gender and control for publication in non-T5 journals and a vector of characteristics $\mathbf{X}$ and $\overline{\mathbf{C}}$.⁴⁶ The marginal effect for gender (indicator for male) is 0.019 (SE 0.038; $p=0.607$) for tenure by the seventh year of tenure-track experience, and -0.045(SE=0.033; $p=0.175$) for tenure during first spell of tenure-track employment. Both estimates are statistically insignificant

⁴⁵These probabilities are computed as proportions of individuals of a certain gender with a certain number of T5 publications who receive tenure

⁴⁶See Equation TA-1 for exact specification